

Doom PWAD QA Report

Run d344176a-3982-4cdd-8c40-10847f70df0f | Map E1M1 | Outcome timeout

Executive Summary

This is an autonomous QA test of E1M1 using an LLM-directed test harness. The test was designed to evaluate map progression, combat, and resource balance, resulting in a timeout after 44 seconds of execution.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: PASS secret_coverage: FAIL
overall_verdict: FAIL

- The automated playthrough did not move to at least 3 distinct position clusters before reaching the timeout.
- The kill/spawned_enemy_count ratio (0.04) is below the 0.5 pass threshold.
- The selected skill summary meets the required health (0.19) and ammo (0.18) ratio thresholds for success.
- Static analysis identifies 8 secret sectors, but 0 were discovered during the run.

No technical problems encountered during this test run.

Test Item And Environment

The test evaluated map E1M1 using the Freedoom1 IWAD on skill difficulty 3. The map contains 69 enemies (including Zombiemen and Imps), 78 items, 3 keys, 8 secret sectors, and measures 5952x3712 units. Static analysis indicates a high-complexity, hitscanner-heavy layout.

The test environment utilized ViZDoom, the gemini-3.1-flash-lite LLM model, and the MCP interface. The test was conducted at difficulty level 3 with a 220-tick maximum duration.

Metric	Value	Metric	Value
Status	completed	Duration	44
Actions	2	LLM calls	2
Final HP	100	Kills	1
Position samples	3	Movement units	419.1
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/d344176a-3982-4cdd-8c40-10847f70df0f.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 3	Notes
Enemies	69	25	44 hidden by skill/multiplayer flags
Items	78	71	7 hidden by skill/multiplayer flags
Secrets	8	8	Sector special 9
Geometry	571 linedefs	99 sectors	5952 x 3712 units

Selected-difficulty estimate: hard. Health ratio: 0.1909. Ammo ratio: 0.1848.

Coverage Evaluation

The automated playthrough tested only a small fraction of the map sectors near the player spawn. Only one enemy was engaged and defeated, and no secret sectors were discovered. Coverage was insufficient for a full assessment of the map's later progression.

Objectives Covered

- **Engaged initial Zombieman:**

Objectives Omitted

- **Navigate the entire map:** The run reached a timeout while the player was still in the starting region.
- **Find secrets:** 0 secrets found despite 8 secret sectors in static analysis; the automated playthrough did not reach the relevant areas.
- **Stress-test geometry:** Limited movement restricted interaction with map-wide features.

Uncovered attributes: The playthrough did not reach doors, lifts, or switches beyond the immediate starting area. Consequently, map-specific scripting and geometry navigation could not be verified.

Defects And Risks

Two minor defects were detected during this run, both classified as severity 3. The most impactful issue is the difficulty-specific spawn mismatch that limits the intended enemy variety and items at the current skill level.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
3	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 3, 44 of 69 enemies and 7 of 78 items from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.
3	3	unreachable_secret	218	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Spawn Area:** medium High density of hitscanners combined with difficulty-specific spawn flags creates an inconsistent combat experience.

Activity Log Summary

1) Static analysis phase (Pre-run), 2) Map exploration phase (ticks 18-218), 3) Combat phase (tick 217), 4) Report generation phase (tick 218).

Exploration was effectively halted by the timeout, preventing the automated playthrough from proceeding to the main combat sequences.

Tick	Event	Position	HP	MCP Action	Stop reason
217	kill	1326.2, -3319.3	100	set_objective	objective_set

Limitations And Recommendations

Locked doors, teleporters, and crusher traps were not encountered as the automated playthrough did not progress beyond the starting area.

Good Quality Areas

- **Start point:** The initial area contains reachable essential equipment, such as the shotgun, to aid early survival.